

Riley Beenders

OT NETWORKS | INDUSTRIAL AUTOMATION & CONTROLS | MULTIDISCIPLINE TECHNICAL PROJECTS

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PROFESSIONAL SUMMARY

Engineering leader with 10+ years of multidisciplinary experience spanning industrial automation, manufacturing controls, and network infrastructure. Builds networked control systems with VLAN segmentation, firewall routing, and documented safety, operational, and software risk analysis. Leads multidiscipline technical projects from design standards, technical drawings, and electrical schematics through deployment, reliability, and field support, and communicates technical requirements to engineers, vendors, operators, and customers.

CORE EXPERTISE

Network & Security: Network infrastructure and design; firewall and reverse-proxy routing; server network and VLAN configuration; segmented industrial device networks; DNS and subdomains; domain-based application access; wireless deployments; cloud gateways

Industrial Automation: Industrial automation and manufacturing controls; Modbus; CAN bus; Allen-Bradley PLCs; LabVIEW data acquisition; sensor and I/O integration; remote operations and diagnostics; motion safety

Systems & Tools: Windows/Linux systems administration; Proxmox virtualization; Docker and Kubernetes; Python; PowerShell; Git/GitHub; SHA-256 integrity verification; data logging and process analysis

Standards & Delivery: Security, operational, and software risk analysis; EU and U.S. standards; 2006/42/EC machinery directive; technical drawings and electrical schematics; technical writing; process standardization; multidiscipline project leadership

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.
- Built an automated LabVIEW QC system that negotiates Modbus communication with three tri-axial laser sensors over a dedicated private VLAN, collecting nine data points (three per laser) every 17.5 ms and charting live tolerances.
- Automated QC reporting: each run generates a formatted PDF stored on a remote server under an ID that becomes a QR code on the product, making field issues traceable.
- Managed and expanded the division's IT infrastructure after the Proteor acquisition, deploying cloud gateways and standardizing software and internal processes to align with company-wide systems.
- Implemented ProteorPrint OS with external software partner SimplyPrint, contributing machine requirements and integration for remote operation, live video, AI-assisted diagnostics, and material, waste, and outcome metrics.
- Led the design and development of ICARUS-Lite, owning its mechanical, electrical, and software design, including custom AC heaters and an IEEE-style electrical design adaptable to 120/240 VAC.
- Addressed CE Machinery Directive requirements on ICARUS-Lite and completed product changes supporting conformity with 2006/42/EC, while reducing assembly time by 80% compared to previous designs.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.
- Provide cross-functional technical support for the product line, from troubleshooting to after-sales service.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer

January 2023 - January 2025

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while retaining hands-on responsibility for mechanical, electrical, firmware, and software systems.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM for orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing.

- Led the design and launch of Gen3 POSEIDON, combining a 1 m³ build volume, filament-and-pellet dual extrusion, braked NEMA 34 Z drives that prevent uncontrolled motion during power loss, industrial-servo XY motion, and a custom gantry.

Lead Electro-Mechanical Engineer

June 2022 - January 2023

- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging while supporting larger, modular machines.
- Developed independently testable electronics enclosures that could be installed in minutes and coordinated custom harnesses, panel designs, and hardware improvements with vendors.

Industrial Engineer

April 2021 - June 2022

- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.
- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry system for a large-scale statue client, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, support, and training.

Engineering Technician

May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized process parameters while documenting design limitations and opportunities for product improvement.

SELECTED PROJECTS

ExtrusionLine Studio - LabVIEW Metrology | Proteor | Replaced a manual measurement workflow with automated Modbus acquisition from three tri-axial laser sensors on a private VLAN, hands-free startup, live tolerance charts, and QR-traceable PDF reports on a remote server.

PROTEOR Print Integration - Remote Operations Software | Proteor | Single interface for remote operation and diagnosis from any networked device, delivered with an external software partner: live video monitoring, machine-vision AI-assisted failure detection and alerts, and tracking of material usage, waste, and print success or failure for analytics-driven decisions.

Modular CAN-Bus Controls Architecture | Filament Innovations | Self-contained, independently testable electronics enclosures connected over CAN bus, separating electrical build work from final machine integration and cutting installation of tested electronics to minutes.

RB's Auto App Updater - PowerShell / Windows Automation | 2026 | One-line PowerShell install that upgrades every WinGet package through an elevated scheduled task: a non-elevated launcher calls protected worker scripts, admin approval is requested once at setup, per-package failures never stop the run, exit codes map to plain-language statuses, and self-updates install only after SHA-256 verification against a version manifest.

Engineering Operations Platform - Kubernetes ERP/CRM | Filament Innovations | Redundant Kubernetes-hosted system connecting engineering design data to operations: centralized orders, inventory, tasks, projects, and customer interactions, and linked dynamic BOM generation to purchasing.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed 750+ wireless access points across six school buildings; automated setup-data entry for 2,500+ Chromebooks with an Arduino-based tool; coordinated distribution with school administrators.

Blue Mountain Ski Resort - Ski Lift Operator | November 2019 - January 2021 | Operated passenger lifts, performed motor and control safety checks, and supported the maintenance team during lift breakdowns while assisting thousands of guests.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Certifications: MathWorks MATLAB, Simulink, Simscape, and Machine Learning Onramps (2026); Lean Six Sigma Yellow Belt and White Belt, Educate 360 (2025); Penn State Certificate in Engineering Design and Tools (2023); Lafayette College Mechanical Engineering Certificate (2019)